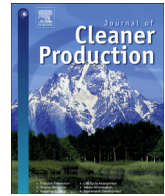




Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Driving forces of solar energy technology innovation and evolution

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ARTICLE INFO

Article history:

Received 13 February 2020

Received in revised form

23 October 2020

Accepted 5 November 2020

Available online xxx

Handling editor: Prof. Jiri Jaromir Klemes

Keywords:

Solar energy technology

Driving forces

Theme evolution

Demand-driven theory

Science advancement

Technology catch-up theory

ABSTRACT

This paper aims to provide an overview of the innovation and evolution of global solar energy technology (SET) and further analyze the driving forces. It addresses the evolution from a longitudinal perspective throughout more than 150 years, and discusses the multiple and interactive driving forces in the process of SET innovation. Our study encompasses three theories, including demand-driven theory, science advancement and technology catch-up theory. We retrieve patent data from PatSnap, a global patent platform, and obtain 313,530 hits in total worldwide, published from 1867 to 2018. Through employing the functions of cell diagram analysis and geographic territory mapping embedded in PatSnap platform, we conduct a relatively comprehensive analysis of SET evolution. Results disclose that the evolutionary trajectory analysis of SET reveals different characteristics in different stages, which involves spatial, main companies, IPCs and hot topics evolution, etc.. Multiple interactive driving forces play a significant role in various development phases. Finally, we give the research contributions and put forward related policy implications.

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1. Introduction

This paper aims to provide an overview of the innovation and evolution of global solar energy technology (SET) and further analyze the driving forces, including demand-driven theory, science advancement, and technology catch-up theory, from a longitudinal perspective over more than 150 years. It has been proven that solar energy intercepted by the Earth far exceeds the demand for global energy consumption (Parida et al., 2011). As a typical and significant renewable energy source, solar energy has attracted significant attention from research scholars worldwide, especially in recent years. Given the urgency of global environmental issues, solar energy is assumed to be best alternative and most promising development direction for future energy prospects (Kannan and Vakeesan, 2016). Compared with other renewable energy sources, it is superior with respect to its cost effectiveness, cleanliness, accessibility, capacity, availability, and efficiency (Kannan and Vakeesan, 2016).

The basis of these valuable discussions and applications stems from steady progress in SET. To resolve the energy crisis, scholars

have discussed issues such as energy demand, the world energy scenario, and the potential applications and barriers of SET for the future (Kabir et al., 2018; Qandile and Sabry, 1998). There is no doubt that the related articles on SET open up new opportunities for the utilization of solar energy resources. However, extant studies have only investigated the evolutionary pathways in a particular region or worldwide (Luan, 2012; Nemet et al., 2017; Urban et al., 2018), and most literature focuses on only one or several specific SET areas, such as solar water heater technology and transparent solar photovoltaic technology (Husain et al., 2018; Kabir et al., 2018; Parida et al., 2011). In addition, they usually highlight the specific application of SET in architecture (Zhou Tao, 2013), water treatment (Malato et al., 2009), or another field or provide heuristic insights into the potential and future prospects of solar energy research (Green and SpringerLink, 2006), leaving an overview analysis of the development characteristics and evolutionary driving forces of the worldwide SET unexplored.

Taking a general view of SET development and pursuing further insight into the driving forces of SET innovation and evolution is critical. Due to the rapid development of SET innovation, the graduated escalation of energy demand, and governments' pursuit of a competitive advantage in this emerging strategic industry, it is important to understand how the technology has evolved over time, covering its origins, spatial, temporal, major enterprise, and

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thematic evolution. Therefore, we can grasp the development trend more easily and scientifically and accelerate the coordinated development of the economy and ecology. More importantly, it is valuable to reveal which driving force is dominant in a particular SET development phase and how multiple driving forces interact.

The remainder of the article is organized as follows. After the introduction, we delineate the theoretical background in the second section. The third part is the research method. The fourth part is the evolution of the SET. We analyze the driving forces in the fifth section, summarize this study, and discuss policy implications in the sixth part.

2. Theoretical background

2.1. Theoretic framework of SET innovation and evolution

We develop this paper following a framework that combines three driving forces: science advancement, demand-driven theory, and technology catch-up theory. The relationships are shown in Fig. 1.

For a long time, the topic of technological innovation has been the focus of attention. As a significant and measurable indicator of research outcomes, patent counts are preferred by scholars in related studies (Johnstone et al., 2010; Benner and Tushman, 2002; Hallonsten and Christensson, 2017; Nevzorova and Karakaya, 2020). Among these works, a large number of investigations have been conducted on thematic evolution, such as learning about the trajectory of electrochemical energy storage, for which Cit-NetExplorer software was used to draw citation chronology charts and study development trends (Wu et al., 2017). However, there are some limitations. This study was well designed and implemented, yet the theme was summarized subjectively, rather than automatically generated by the program. Although there have been some studies on thematic evolution (Sun and Zhai, 2018), the relevant literature only focused on science rather than technology, and the timespan was usually very limited, which resulted in losing an overview of the evolution of the whole domain.

Identifying “core journal literatures and persistent research themes in an emerging interdisciplinary field” (McCain, 2010) can facilitate an understanding of technological thematic evolution. A series of bibliometric indicators, including publications, citations, and collaborations, have also been used to analyze emerging topics and their evolution (González-Alcaide et al., 2018). It is therefore true that methods and indicators can be used interactively within different disciplines and fields. Moreover, mature techniques and methods used in detecting scientific thematic evolution can also be

applied to analyze changes in technological topics of SET research.

2.2. Demand-driven theory and technological innovation

Over the last few years, studies on the economics of technological change have identified demand as one of the most powerful driving forces for innovation (Schmookler, 1962, 1966). Specifically, rising demand leads to an increase in technological innovation input and output of enterprises (González-Alcaide et al., 2018), which promotes both innovation speed and direction.

As a result of innovation, the change in survival demand orientation and scale as well as the profitability of positive expectations, are the effective driving forces for enterprises and researchers to carry out technological innovation. Concurrently, the specificity and potential profitability of technological innovation also increase the demand size or cultivate new demand. The process of invention is an oscillation between demand and technological innovation (Suzuki, 2015).

2.3. Science advancement and technological innovation

Science advancement is a knowledge system of logical interpretation in scientific disciplines, which consists of a series of concepts, judgments, and inferences. The development of science and the innovation of technology are interdependent. As the basis of technological innovation, science advancement promotes technological innovation, and new technological achievements in turn influence science advancement through further technological improvement and development (Rickert and Nagy, 1999; Sanmartin, 1993). This viewpoint can be proven by the technological innovation strategy of the United States, which requires that basic science advancement should maintain its world-leading position in research and development (Damanpour, 1991). Because the global leading level of scientific progress is highly correlated with subsequent technological development, it triggers related technological innovation (Makri et al., 2006). For example, when technology makes great progress, it will strengthen the United States' leading status of scientific research. In other words, once the technology is applied to the results of science advancement, it will evolve in parallel (Coccia, 2017). With the continuous growth of scientific knowledge, continuous science advancement is called “scientific progress.” Moreover, technological progress is known as “technological evolution,” which is the result of replacing the existing technology with new and better ones. Consequently, along with rapid scientific advancement, technology has also been promoted quickly. At present, human society has entered an unprecedented era of innovation intensification. Breakthroughs in science advancement have become the source of technological innovation in all professions and trades (Zhou et al., 2016).

2.4. Technology catch-up theory and technological innovation

At the macro level, technology catch-up theory refers to the technological gap between technical entities with relatively low technology levels in late-developing countries and advanced levels in developed countries. Over the past few years, several late-developing countries have reduced or eliminated this gap. The strategic guiding ideology of technology catch-up theory is positioned in the non-equilibrium conception that the strategic goal transcends resource endowment, such as narrow regions (e.g., Singapore) with a global layout. The catch-up effect focuses mainly on leading sectors or industries (especially solar energy) with sustainable innovation capabilities. Through industrial radiation and market induction, they create and expand distinctive and influential resources, rapidly develop economic capacity, and

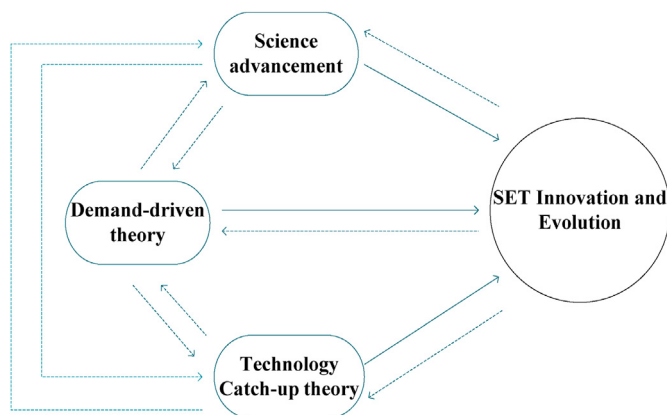


Fig. 1. Theoretical framework.

further promote the matching and development of other sectors or industries (González-González et al., 2014). For example, companies in the US, Europe, and Japan lead the way in technology-intensive industries. These countries are also recognized as giants in the global technology field because of their high level of technical capabilities. In contrast, companies in emerging Asian economies are lagging in most technology sectors. However, in recent years, Chinese and South Korean companies have rapidly improved their technology capabilities. In some fields or industries, they have pursued or even surpassed existing companies in developed countries with the help of institutional conditions, such as innovation promotion policy (Furman and Hayes, 2004; Hu and Mathews, 2005; Park and Lee, 2006; Wang and Tsai, 2010) and establishment of scientific institutions (Bernardes A T, 2003). This can be confirmed by the shift of the technical center through the patent records of the US Patent and Trademark Office (Gao, 2011).

Regarding micro-level technological advancement, leading enterprises in developed countries have established relevant technical paradigms, with sufficient resources, technical foundations, design rules, and customer lock-in. However, this situation highlights the problem of creating technical barriers for late-developing countries, which have difficulty in acquiring technology and promoting innovation quickly. In addition, the limited R&D investment and long development cycle can easily hinder rapid technology leapfrogging. These factors force latecomers to adopt the strategic and competitive decision of technology catch-up theory when the window of opportunity opens. The choice is considered from two aspects. On the one hand, late-developing countries must consider how to build technical capabilities through learning innovation knowledge, prioritizing innovation capabilities from the beginning to ensure competitiveness (Fan, 2006), sharing complementary resources, and creating synergies with R&D cooperation (Lee et al., 2010). On the other hand, it emphasizes the choice of corporate technology catch-up strategy and the influence of corporate organizational factors. In this way, we find that many technological innovations are externally promoted (Miao et al., 2018).

2.5. Interaction among three theories and technological innovation

Under certain demands, the development of science advancement and technology catch-up can promote each other to push the continuous development of new technology, such as SET; however, due to the limitation of demand, this interaction may soon reach a threshold. Therefore, after a certain period, the demand force can in turn reduce the speed of science advancement and the power of technology catch-up, which produces a negative interaction between the two forces and severely hinders the development of new technology.

In addition, when the value of science advancement is fixed at a certain level, technology catch-up can lose its direction due to the lack of fundamental innovation. Moreover, new technology may be application-oriented, repetitious, and horizontal, which, of course, will lead to bottlenecks in technology catch-up and hinder its continuous evolution. Similarly, demand also faces bottlenecks and hovers at a level when researchers cannot make breakthroughs in key technical issues. Even if the development of new technology continues to promote demand, it cannot meet the corresponding standards.

Finally, at a certain level of technological catch-up, the promotion of science advancement and vigorous demand can be realized simultaneously. For example, the demand for upgrading is supported by the corresponding scientific advancement. Simultaneously, science advancement is strongly driven by demand; however, the interaction between the two forces also suffers the ceiling effect owing to a limited willingness of enterprises and

governments to promote the development of new technology. Therefore, the advancement in new technology is limited. However, with the support of these three forces, technology development will inevitably follow.

3. Methods

3.1. Model specification

Patent publications are extensively used in exploring technological innovation and evolution (Jee et al., 2019; Karvonen et al., 2016). They contain rich technical information and innovative results and are usually considered the proper dataSET for analyzing technology evolution (Awate and Mudambi, 2018). Therefore, in this study, SET patent data were selected to trace the evolution of this field (Fig. 2).

After the search, refinement, and method description of SET data, we analyzed the evolution of SET, including spatial, temporal, major enterprise, and thematic evolution. Finally, we discuss the multiple driving forces in SET innovation and evolution. Text clustering analysis, which relies on the Lingo3G text document clustering algorithm (Corner, 2019) is embedded in the PatSnap database (PatSnap, 2019a) and was used to trace the evolutionary trajectory of global SET from 1867 to 2018 for more than 150 years. Moreover, the geographic territory mapping function embedded in the PatSnap database (PatSnap, 2019b) was used to analyze the spatial evolution. The geographic classification shows the percentage breakdown of patents in the technical field across different authorities (PatSnap, 2019a). It is helpful for us to understand the regional markets in which the technology field is prominent and commercialized.

3.2. Data collection

Patent data of global SET in this study have been retrieved from the PatSnap patent database, including more than 133 million patents from 107 patent authorities worldwide since 1790 (PatSnap, 2019b). The search strategy operates according to the definition of SET provided by the World Intellectual Property Organization (WIPO) (Luan et al., 2013; WIPO, 2019), with a group of IPC symbols listed in Fig. 2. The IPC-based definition of the SET employed here can capture the vast majority of patent filings in this area because the correspondence between IPC symbols and technology fields is not always clear-cut, that is, there is no one-to-one relationship.

The first three steps in Fig. 2 focus on searching and refining the

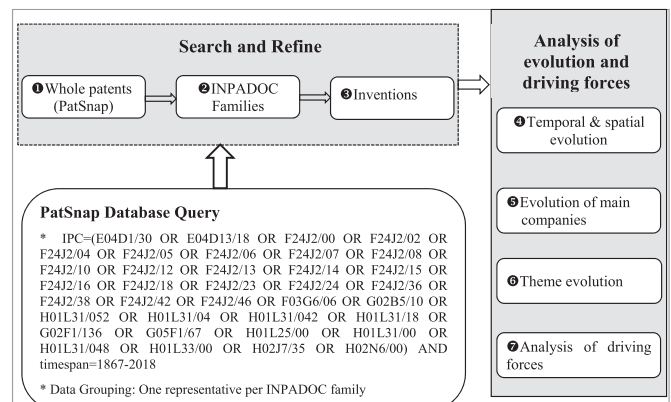


Fig. 2. Model specification.

patent data of SET.

- In step 1, IPC symbols listed in the diagram were employed to capture all patents in all study years, and **772,174** hits were attained in total.
- In step 2, all patents were refined using the INPADOC strategy (Kulp, 1984; Rubitschka, 1982), that is, only the first patent of each International Patent Documentation (INPADOC) was reselected in such a way as to avoid redundant data for the same invention (de Paulo and Porto, 2018), and the selection was narrowed to **400,957** hits.
- In step 3, invention patents were further collected and refined, excluding data of utility patents or design patents. Finally, **313,530** hits were collected and applied as the SET patent dataset in this study.

3.3. Visualization scheme

This study attempted to determine the central topics in SET at each stage and identify their changes over time. The cell diagram analysis function embedded in the PatSnap platform (PatSnap, 2019a) shows the hot topics of patents belonging to the top organizations in a particular technology area. The relative coverage is represented by the number of cells under each organization, and each cell represents the same number of patents. Such an analysis allows us to better understand the topics that are relevant to the top-level organizations in a given technology domain at a particular stage by identifying the distinctions between the technological focus of each top-level organization and providing a visual overview map in our mind.

4. SET evolution

4.1. Origin and evolution of SET

4.1.1. Pre-patenting era: SET origins

The utilization of SET can be traced back to ancient times, when people built their shelters orienting to the sun, designed natural ventilation spaces, used the sun to dry items, and selected materials from natural materials with favorable thermal mass or light dispersing qualities. In 1615, Salomon de Caus, a French engineer, published a book showing a steam-driven pump that utilized a solar heating system to expand air. In 1774, Lavoisier, the well-known French chemist, made a powerful lens to concentrate solar radiation (Kalogirou, 2014). During the 19th century, in Europe, North Africa, and the United States, people attempted to convert solar energy into other forms of energy by operating steam engines. In 1839, Edmond Becquerel, a French physicist, discovered the photovoltaic effect, which is the operating principle of the solar cell (Kalogirou, 2014).

4.1.2. Phases of solar energy patenting

The development trend of solar energy patents published from 1867 to 2018 is demonstrated in Fig. 3. To trace the evolution of SET over such a long history, the curve of solar energy patents was further divided into four stages. These stages were separated according to perceived shifts in annual patents. Combined with the knee of the curve and the threshold, the identification of the inflection point depends on the observation of the visual curve, and the threshold was determined by checking the data year by year, as shown in Table 1.

According to the changing characteristics of the curve and threshold at each stage, we can name each of the four phases as follows: Phase I: 1867–1919, infancy stage; Phase II: 1920–1974,

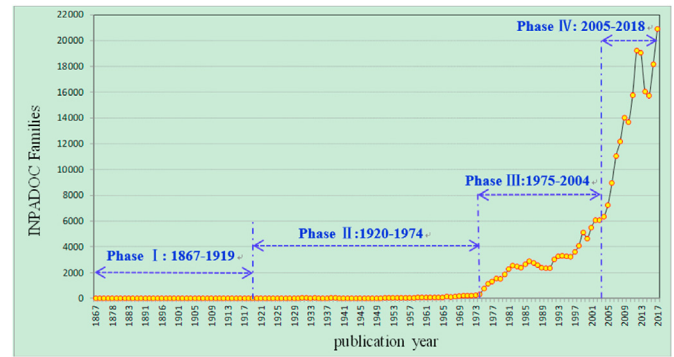


Fig. 3. Development trend and four phases of solar energy patents: 1867–2018. Source: PatSnap database.

Table 1

Stage thresholds.

	1867–1919	1920–1974	1975–2004	2005–2018
Threshold	$n < 10$	$10 < n < 500$	$500 < n < 7000$	$n > 7000$

Note: The threshold, n , is the number of patents in each year.

exploration stage; Phase III: 1975–2004, rapid development stage; and Phase IV: 2005–2018, high-speed development stage.

Fig. 3 shows that the development speed of solar energy was very slow in the initial phase. In the first stage, from 1867 to 1919, there were less than 10 patents per year. In the long exploration stage from 1920 to 1974, the number of patents per year ranged from dozens to hundreds, and during that time, the highest number of patents annually was less than 500. It began to grow faster from 1975 to 2004. After 2005, the speed of development has been greatly accelerated, and there were more than 100,000 patents every year until 2018.

4.2. Spatial evolution

The spatial evolution analysis provides a holistic and comprehensive understanding of the origin, development, and transfer of world SET centers from a longitudinal perspective.

4.2.1. Evolution of SET patenting in top authorities by year

Analyzing the 150-year span from 1867 to 2018, this study explored which authorities exhibit a long engagement with SET and identified emerging countries and regions. The annual distribution of SET patenting during 1867–2018 is shown in Fig. 4.

Fig. 4 reveals the annual dynamic development trend of SET

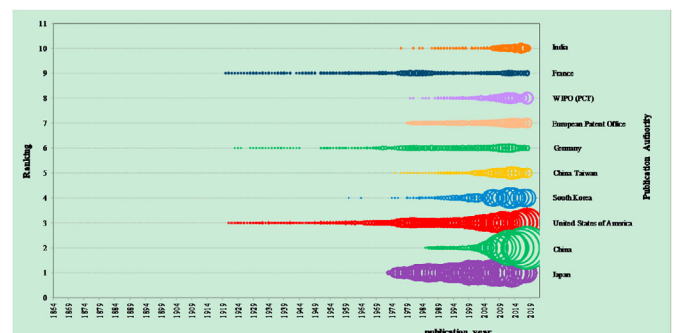


Fig. 4. Evolution of SET patenting in top 10 authorities by year: 1867–2018.

patenting in the top 10 authorities from 1867 to 2018. From an overview, the United States, France, and Germany have taken longer to file SET patents than other authorities. China, India, and South Korea are apparently emerging countries of SET patent applications. In the exploration stage of 1920–1974, the United States, France, and Germany were outstanding in SET patenting. In the rapid development stage of 1975–2004, Japan, the United States, South Korea, China, and Germany were salient in terms of SET patenting activities. In the high-speed development from 2005 to 2018, China was the most outstanding authority with the most SET patents, followed by Japan, the United States, and South Korea.

4.2.2. Shifting of SET patenting center

The shifting of the world science center has long attracted scholars' attention. Bernal first proposed the transfer of the world science center (Bernal, 1954); Yuasa expounded the phenomenon of the shifting of the world science center according to the share of a specific country in the global total, with respect to scientific achievements (Yuasa, 1962). If the share of a certain country was over 25%, then the country was the world science center. Zhao and Jiang further revealed the phenomenon of global science centers based on data of chronological events in the natural sciences (Zhao et al., 1985). Using the geographic territory mapping function of the Patsnap database, we obtained four geographic maps matching the

four development phases, demonstrating the percentage of SET patenting, as shown in Fig. 5.

Fig. 5 reveals the shifting of SET patent application activity centers worldwide: the United States (1867–1919 and 1920–1974), Japan (1975–2004), and China (2005–2018). Table 2 lists the ratio of SET patents in total patents in each country/region in each phase.

Table 2 reveals the shifting of the SET patenting center more specifically, illustrating a change from the high concentration in the early stage of 1867–1919, to flowering widely across the world in the present time. In the first stage of 1867–1919, the United States is far ahead of other authorities with a ratio of over 70%. The concentration degree decreases gradually over time; taking the United States as an example: 71.65% (1867–1919), 43.73% (1920–1974), 18.53% (1975–2004), and 17.97% (2005–2018). The percentage ratio of the top authority decreases over time: the United States with a ratio of 71.65% (1867–1919), the United States with a ratio of 43.73% (1920–1974), Japan with a ratio of 56.85% (1975–2004), and China with a ratio of 34.47% (2005–2018).

4.3. Evolution of main companies

Industry is the mainstream of technological innovation (Laanti et al., 2007). Where gathers more leading and innovative companies, where can be the innovation center. Before 1919, the characteristics of SET patenting tended to be individuals rather than companies. Table 3 lists the main companies in each phase;

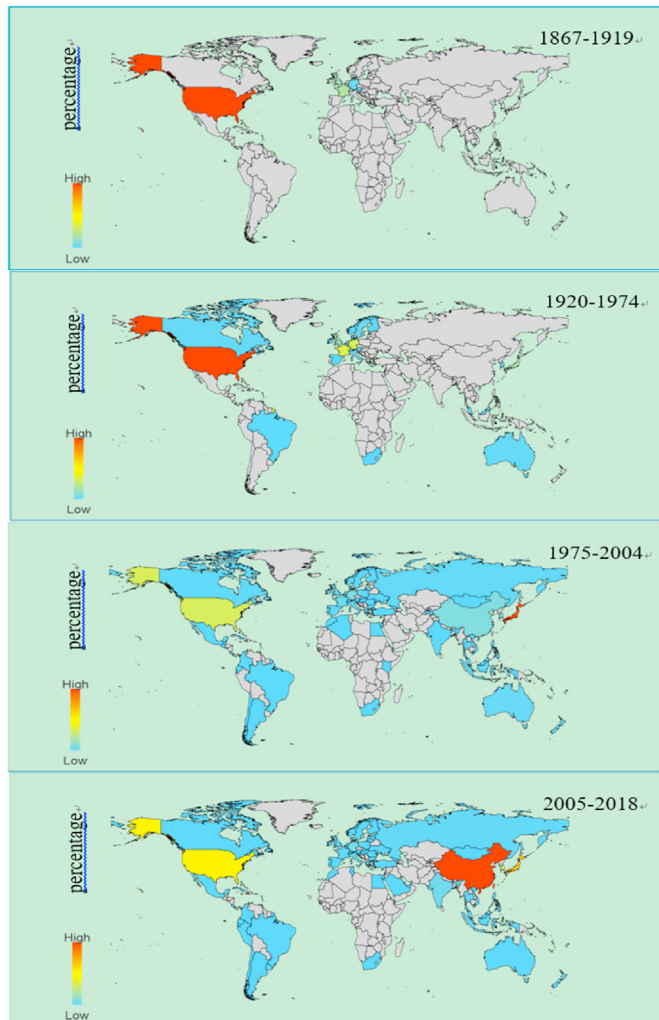


Fig. 5. Percentage breakdown of SET patenting across different authorities.

Table 2

Ratio of SET patents to total patents in each country/region in each phase: 1867–2018.

rank	authority	ratio
1867–1919		
1	United States	71.65%
2	France	14.17%
3	United Kingdom	11.81%
4	Germany	1.57%
5	Austria	0.79%
1920–1974		
1	United States	43.73%
2	France	15.53%
3	Germany	14.85%
4	Japan	8.50%
5	United Kingdom	6.85%
6	Switzerland	2.18%
7	Soviet Union	1.68%
8	Netherlands	1.56%
9	Sweden	1.11%
10	Canada	0.87%
1975–2004		
1	Japan	56.85%
2	United States	18.53%
3	South Korea	4.04%
4	China	4.02%
5	Germany	3.82%
6	European Patent Office	2.00%
7	Soviet Union/Russia	1.53%
8	France	1.45%
9	Australia	1.15%
10	China Taiwan	1.09%
2005–2018		
1	China	34.47%
2	Japan	22.29%
3	United States	17.97%
4	South Korea	10.50%
5	China Taiwan	3.54%
6	WIPO (PCT)	2.47%
7	European Patent Office	2.25%
8	Germany	1.69%
9	India	1.29%
10	Russia	0.47%

Table 3
Main companies in the latter three phases: 1920–2018.

rank	Standardized Current Assignee	INPADOC Families	Share	Region
Phase II: 1920–1974				
1	Siemens	145	4.18%	Germany
2	General Electric	98	2.83%	United States
3	Bell Labs	85	2.45%	United States
4	IBM	71	2.05%	United States
5	Philips	55	1.59%	Netherlands
6	RCA	37	1.07%	United States
7	Westinghouse Electric	36	1.04%	United States
8	North American Philips	36	1.04%	United States
9	Texas Instruments	36	1.04%	United States
10	Radio Corp of America	32	0.92%	United States
Phase III: 1975–2004				
1	Panasonic	5007	4.73%	Japan
2	Sharp Corporation	4523	4.27%	Japan
3	Toshiba	4196	3.96%	Japan
4	Sanyo	3049	2.88%	Japan
5	Seiko Epson	2952	2.79%	Japan
6	Canon Kaushiki Kaisha	2928	2.76%	Japan
7	Hitachi	2753	2.60%	Japan
8	NEC	2525	2.38%	Japan
9	Mitsubishi Electric	2346	2.21%	Japan
10	Fujitsu	2186	2.06%	Japan
Phase IV: 2005–2018				
1	Sharp Corporation	6064	2.79%	Japan
2	LG Display	4901	2.25%	South Korea
3	Samsung Display Devices	4007	1.84%	South Korea
4	Samsung Electronics	3749	1.72%	South Korea
5	Semiconductor Energy Laboratory	3197	1.47%	Japan
6	Seiko Epson	2763	1.27%	Japan
7	Panasonic	2606	1.20%	Japan
8	BOE Tech Group	2479	1.14%	China
9	AU Optronics	2189	1.01%	China Taiwan
10	Mitsubishi Electric	2163	0.99%	Japan

specifically, the bubble diagram of the top 5 companies in the stage of 1920–1974 is visualized to present a clear view of the early patenting activities of corporations. Theoretically, enterprises always try to establish their sustainable competitive advantage through SET innovation. This behavior requires them to create and explore new demands to obtain sufficient profits; in this way, they can invest more resources in related R&D activities. This means that the evolution and ranking of main companies will determine the ability to achieve this goal. Companies, especially the top ones, have their own laboratories, but also contribute to the innovation of the SET industry. Due to the complexity of technology, they must connect to universities and other research institutions to increase the effectiveness of innovation and decrease various risks. This will indirectly promote the basic research and progress of SET. Finally, technology catch-up companies can accumulate more knowledge and R&D experience and form more innovation networks, such as strategic alliances, with a change in their ranking. On the contrary, the evolution of main companies stimulates further SET innovation.

4.3.1. Main companies in each phase

The standardization of the current assignee name is chosen as the main company's name. The top 10 companies in the last 3 development stages: 1920–1974, 1975–2004, and 2005–2018, are listed in Table 3.

Table 3 reveals the evolution of central companies over time. In the stage of 1920–1974, 8 of the 10 top companies were from the United States, and 2 companies were from Germany and the Netherlands, indicating that the SET patenting center was located in the United States during this period. In the stage of 1975–2004, all top 10 companies were from Japan, revealing that the SET patenting center was located in Japan during this period. In the stage of 2005–2018, the top 10 companies appear to be more diverse in

nationality, covering the authorities of Japan, South Korea, China, and 5 of the top 10 companies at this stage were from Japan.

4.3.2. Evolution of top 5 companies in 1920–1974 by year

As mentioned earlier, prior to 1919, applications for SET patents were usually filed by individuals rather than corporations. To identify early adopters and examine their SET patenting activities in detail, we created a single bubble diagram for the top 5 companies in the stage of 1920–1974, as shown in Fig. 6.

Fig. 6 reveals that 3 of the top 5 companies were from the United States, and the other 2 companies were from German and the Netherlands. Siemens and General Electric were the very early companies engaging in SET patenting. SET patenting activity was very active around 1954; for example, the world's first solar cell was made from single-crystal silicon by a scientist in Bell Labs in 1954, with a photoelectric conversion rate of approximately 6%

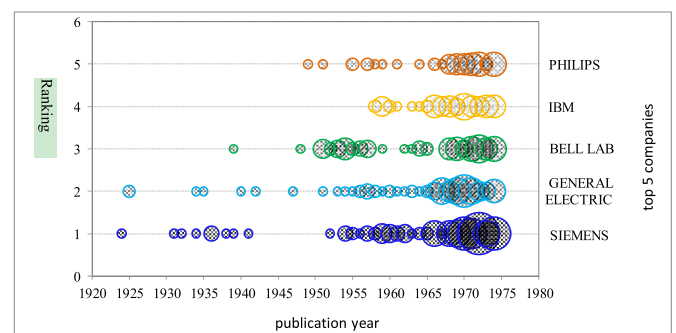


Fig. 6. Evolution of top 5 companies in 1920–1974.

(Goetzberger and Hebling, 2000). Fig. 6 also demonstrates that SET patenting activities of the top 5 companies were extremely active around 1973. This is because the first energy crisis took place in the early the 1970s, which promoted the industry's search for sustainable and alternative energy (Shaahid and Elhadidy, 2004; Youm, 2000) and led to a shift from emphasis on fossil energy to solar energy (Nelson, 2003).

4.4. Thematic evolution

4.4.1. Evolution of IPCs

Based on the Strasbourg agreement on *international patent classification* (IPC) in 1971, the IPC system covers eight sections in total. That is, Section A - Human Necessities; Section B - Performing Operations/Transporting; Section C - Chemistry/Metallurgy; Section D - Textiles/Paper; Section E - Fixed Constructions; Section F - Mechanical Engineering/Lighting/Heating/Weapons/Blasting; and Section G - Physics; and Section H - Electricity. Analyzing the share of each IPC section in different stages can help us understand the dominant sections in a specific development phase. The evolution of eight IPC sections over time in various stages is shown in Fig. 7.

Fig. 7 shows the proportion for each IPC section in each stage. The most significant IPC section in the early period of 1867–1919 is section F, with the technology definition of mechanical engineering, lighting, heating, weapons, and blasting and shows that lighting and heating mechanical engineering is the focus of SET in this initial phase. Section H, with the subject of electricity, wins the first position by a landslide in the following three stages, and its share rises gradually from 67.57% to 72.91%–79.09%, indicating that almost 80% patents involve in electricity in the period of 2005–2018. Moreover, the share of patents pertinent to section F appears to have a downward trend from 81.10% to 22.84%, 18.14%, and finally to 16.88%.

4.4.2. Evolution of central topics

According to the method of the visualization scheme, the central topics of SET patenting in various phases from 1867 to 2018 are demonstrated in Fig. 8.

Fig. 8 reveals the topic evolution of the SET over time. Limited to science advancement, mechanical engineering technology on utilizing solar light, solar heat, and solar heaters has become the focus of attention in the initial phase of 1867–1919. With the scientific discovery of the photovoltaic effect, the operating principle of the solar cell in 1839, and the application of solar energy in the early 1920s, the electricity has led to the development of SET. Specifically, semiconductor device technology remained a vibrant scientific

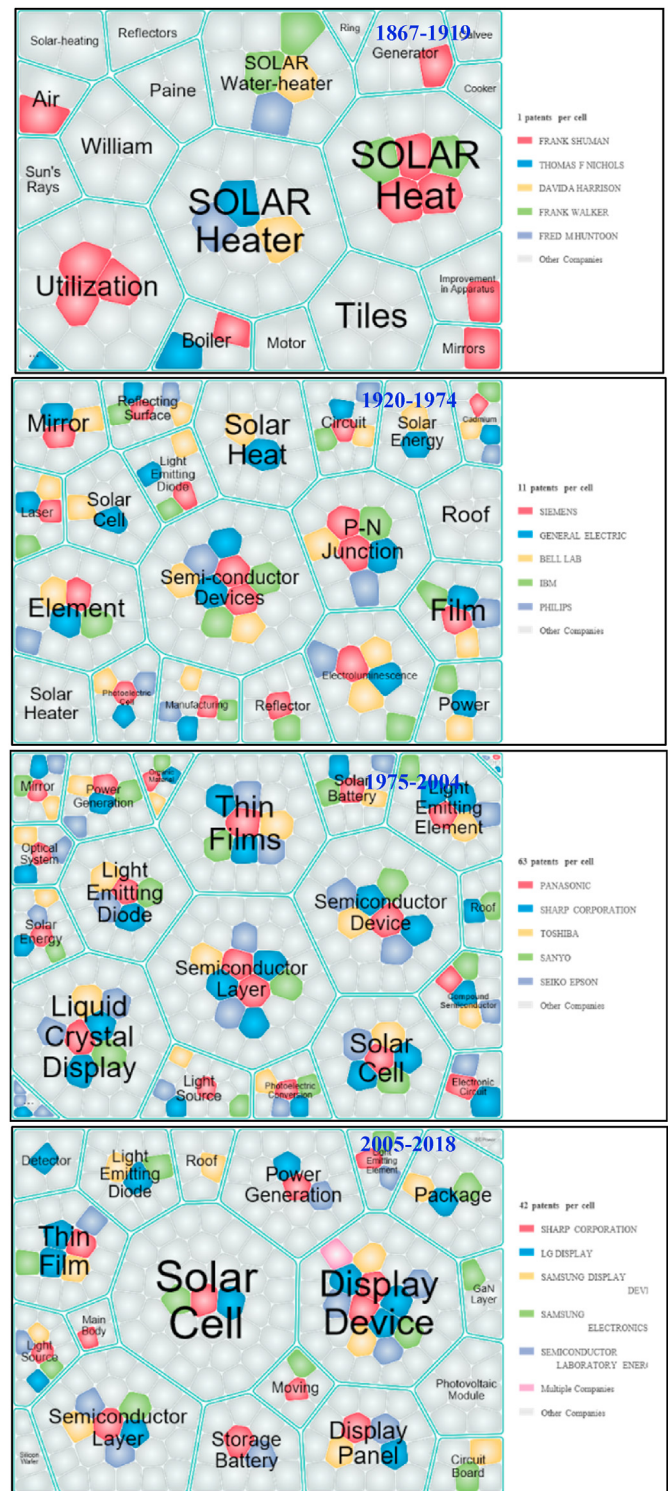


Fig. 8. Hot topics of SET patenting in various phases: 1867–2018.

focus during the phases 1920–1974 and 1975–2004. During these phases, environmental protection issues have been raised, and technology has been developed. As a result, there has been unprecedented development in clean energy-related technology, including solar cells, solar power generation, storage batteries, and light-emitting diodes.

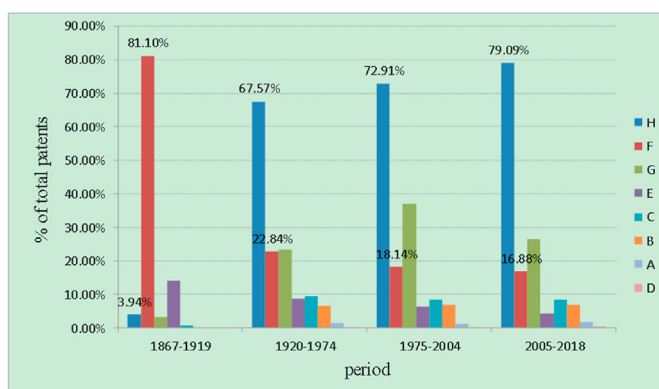


Fig. 7. Evolution of 8 IPC sections in various periods.

5. Analysis of forces driving SET innovation and evolution

The evolutionary trajectory analysis of SET reveals different characteristics in different stages, including spatial, main companies, IPCs, and hot topic evolution. We found that the interactivity of driving forces, including *demand*, *science advancement*, and *technology catch-up*, not only play a crucial role in the innovation and evolution of SET, but also give rise to varied effects at different stages.

5.1. Demand

Demand encompasses *survival and living needs* in the ancient era, *development demand* in the oil crisis era, and the *demand for sustainable development and environmental protection*.

5.1.1. Survival and living needs

In the initial era, survival and living needs were the most significant driving forces for human utilization of solar energy. For example, to obtain heat and keep warm, dry food, and maintain long-term storage, solar energy is being utilized by humans unconsciously in simple and unsophisticated ways, such as orienting shelters in the direction of the sun to get light and heat. From 1867 to 1919, the IPCs and topics mainly focused on phrases regarding the utilization of solar light, solar heat, solar water heaters, boilers, and reflectors. The United States and European nations, including France, Germany, and the United Kingdom, were the cradles of SET due to their contribution to the development of industry and related technologies in the pre-patenting and early patenting eras.

5.1.2. Development demand

The scarcity and unbalanced distribution of fossil energy were gradually recognized with the development of industry, especially the two oil crises in 1973 and 1979. Since then, the search for diversified, alternative, renewable, and green energy has become the focus of global attention. Due to the lack of natural energy in its own land, Japan first implemented a new energy development strategy, the *Japan Sunshine Project*, to ensure energy security and industry development. A series of policies and a huge amount of investment in the exploitation of clean energy have been made by the Japanese government, which has made considerable progress in SET during the past 30 years, from 1975 to 2004. As shown in Table 2, Japan ranked first with a share of 56.85% during this period. China, South Korea, and other Asian countries have also realized the significance of the R&D of SET and took measures to accelerate the R&D investment in SET.

5.1.3. Demand of sustainable development and environmental protection

Industrial boom and economic growth are significant, yet are they most important for humans and society? Issues regarding the threat of energy depletion and serious air pollution that accompany boom and growth make us ponder this. With the proposal of the theory of sustainable development and the improvement of environmental protection consciousness, it has become the best and most necessary choice for humans to seek a type of inexhaustible and green energy. Under the requirements of the Horizon 2020 climate and energy package, EU member states must reduce greenhouse gas emissions by 20%, increase the proportion of renewable energy to 20%, and improve energy efficiency by 20% by 2020 (Szilárd et al., 1951). This demand for energy production and consumption largely influences the utilization of solar energy, which leads to an increase in SET patenting activities.

5.2. Science advancement

Science advancement is another key driving force boosting the progress of SET, such as the discovery of the photovoltaic effect in 1839, the disclosing of the semiconductor photoelectric effect in 1873, and the contribution of the Israel scientist Tabor to the spectral selectivity coatings in 1955. These scientific advances have greatly boosted the improvement and development of SET, attracted the application of SET-related technologies, and established the characteristics of patent applications in different stages. In particular, the latter two breakthroughs have laid a technical foundation for solar energy utilization to enter the modern development period. For example, thermophotovoltaic cells and their applications are mainly based on the thermophotovoltaic effect theory, which promoted the US Patent Office to file 331 patents from 1980 to 2019 (before July).

Specifically, at the firm level, we found that early prolific innovative companies in SET before 1974 mainly came from the United States, Germany, and the Netherlands, such as Siemens, General Electric, Bell Labs, IBM, and Philips, as shown in Table 3. Scientific studies on SET's industrial laboratories have accelerated the pace of solar energy progress. For example, Bell Labs invented the first solar cell in 1941 and obtained a monumental advance in SET. The works can be broadly divided into three categories: basic research, systems engineering, and application development. AT&T gradually developed a highly structured R&D sector, which brought respect to businessmen for science and technology to a new altitude. Subsequently, the spectral selectivity coatings, a solar cell based on aluminum-gallium-arsenide developed by IBM greatly improved the conversion rate. In sum, before 1975, science advancement and technological innovation of enterprises in the SET research area, especially those in Europe and the United States, have laid a significant foundation for the subsequent development of SET. With the increasing emphasis on new energy after the energy crisis in the early the 1970s, an increasing number of enterprises have established their own R&D facilities, which has gradually become the mainstream of SET innovation. In the 1970s, to meet development needs and seek alternative energy, a large number of R&D companies in SET emerged in Japan, and they filed a large number of SET patents and promoted SET significantly. In the 1980s, to meet the needs of sustainable development and environmental protection, governments around the world focused on the R&D of new energy technologies, which has led to the emergence of new companies in SET, especially in China and South Korea.

5.3. Technology catch-up

Since SET was originally developed by Europe and the United States, its related industries are gradually regarded as strategic, knowledge-intensive, and emerging industries, which can effectively propel economic growth. Therefore, the huge benefits of SET have attracted the interest of latecomers. Governments and enterprises in Asia, especially China and South Korea, invest in this technology as an effective way to create advantages in the future. The ambition of governments and enterprises proved to be a catalyst for the prosperous creation of new SET and the surging application of patents. Taking the government of China as an example, "The 13th Five-year Plan," "National Key R&D Program," and many subsidies and tax policies have been designed and implemented to promote the development of the solar photovoltaic industry. Technology-related factors play a significant role in their catch-up, rather than the support of the domestic market (Quitrow et al., 2017). Through unremitting innovation efforts, they have climbed from the bottom to the top of the supply chain and mastered cutting-edge technologies during the past 20 years.

5.4. Summary

The development of SET in different stages is driven by different forces alone or in combination. These influential driving forces have promoted the innovation and evolution of SET. These results are consistent with the information provided by SET books (Antolín, E and Martí, A, 2012), journals (e.g. Annual Review of Energy), and websites (e.g. the website of Solar Energy Technologies Office). Specifically, we can attribute the driving forces at each phase (Table 4).

6. Conclusion and policy implications

6.1. Conclusion

Solar energy is one of the most promising clean technologies for future energy production. Disclosing the evolution and driving forces for SET development gives us clear information on how to strongly push the cleaner energy industry. In this study, we presented a series of characteristics of the innovation and evolution of SET and further analyzed the driving forces of SET evolution by selecting longitudinal worldwide patent data from 1867 to 2018. The main research outcomes are as follows, covering two parts: one is the findings derived from the analysis of SET evolution, and the other is the outcomes derived from the analysis of driving forces.

6.1.1. SET evolution findings

● Findings derived from temporal evolution analysis

The development curve of solar energy patents published from 1867 to 2018 can be divided into four phases: the infancy stage from 1867 to 1919, exploration stage from 1920 to 1974, rapid development stage from 1975 to 2004, and high-speed development stage from 2005 to 2018, according to the changing characteristics of the curve and interval threshold (the number of SET patents increases per year): with fewer than 10 patents, between 10 and 500 patents, between 500 and 7000 patents, and more than 7000 patents.

● Findings derived from spatial evolution analysis

SET patenting centers, highly concentrated in the past, are now blossoming all over the world. The analysis of spatial evolution reveals this. In terms of ratio, the shifting of the SET patenting center worldwide changed from the United States (with a ratio of 71.65% from 1867 to 1919, with a ratio of 43.73% during 1920–1974), Japan (with a ratio of 56.85% during 1975–2004), to China (with a ratio of 34.47% during 2005–2018).

● Findings derived from main enterprise evolution analysis

Table 4
Three main forces in different stages.

Year	Main Force		
	D	S	T
1867–1919	▲	▲	○
1920–1974	▲	▲	▲
1975–2004	▲	○	○
2005–2018	▲	○	▲

Note: D, S, and T denote the force of demand-driven, science advancement, and technology catch-up, respectively. ▲ (○) indicates the main (not main) force in focal stage.

Prior to 1919, SET patent applicants were primarily individuals rather than companies. By analyzing the evolution of the major companies in the following three stages, the distribution of the top 10 companies changed from highly centralized to more diversified: 8 prolific companies were from the United States from 1920 to 1974; all 10 productive companies were from Japan during 1975–2004. The top 10 companies were from Japan, South Korea, and China from 2005 to 2018.

● Findings derived from thematic evolution analysis

Analyzing IPC evolution elucidated the shift of the dominant IPC section from F to H over time. The share of section F decreased from 81.10% to 22.84%, 18.14%, and 16.88%, respectively, whereas the share of section H increased from 3.94%, to 67.57%, 72.91%, and 79.09%, respectively. The hot topics ranged from the humble beginnings of solar heat and solar heater, semiconductor devices, semiconductor layers, and solar cells over time. It is notable that topics such as solar energy have obtained significant attention. Solar cells have become the hottest topic among researchers and have made further progress.

By analyzing the evolution of SET in time, space, major enterprises, and themes shows the general characteristics of the evolution of clean energy represented by solar energy.

6.1.2. Three forces driving clean energy development

In our study, the development of SET can be attributed to three driving forces. From the viewpoint of demand, we conclude that the need for human survival and living as well as the pursuit of sustainable development and environmental protection, is the main driving force for SET. For scientific advances, we find that major scientific breakthroughs such as the discovery of the photovoltaic effect and semiconductor photoelectric effect, the invention of the first solar cell, and the contribution of the spectral selectivity coatings has made significant contributions to SET. Finally, due to the competition of expected revenue from the solar energy industry, the implementation of the technology catch-up strategy by governments and companies has become another important driving force for the innovation of SET. In other words, the current evolutionary paths have created a basic development circumstance for SET innovation, and new inventions will continue to be developed. In the future, these processes will accelerate the use of non-solar clean energies drastically as they become more relevant to the three forces.

Our study contributes significantly to contemporary research. First, the overall evolution trend of SET under the interaction of the three driving forces can help us understand the internal dynamics of the most promising and clean SET, especially the differences and connections under the influence of the characteristics of regions and periods. Second, our work provides an overview of which clean SET are lacking, which are over-developed, and their relevance. Based on these observations, we can get an in-depth knowledge of the main incentives or supplements that should be given at different stages to promote the balanced development of clean SET. Third, the solar energy industry will be accompanied by waste pollution. For example, as the lifeline of photovoltaic enterprises, polysilicon produces a large number of by-products of toxic substances during the production process, such as silicon tetrachloride, trichlorosilane, and silicon dichloride. After fully understanding the dynamics of technology evolution highlighted in this paper, the three driving forces can be used to guide the evolution of SET to diversified clean technology. Moreover, by improving the utilization rate of raw materials and reducing the emission of pollutants, the solar energy industry can achieve a closed-loop, sustainable, and clean production process.

6.2. Policy implications

6.2.1. Promoting cleaner energy demand strategy

Cleaner production is an effective tool for realizing sustainable global development. They attach great importance to the research, development, and utilization of SET. At present, there have been a large number of patent applications and approvals; however, new efforts are still needed to promote cleaner energy demand strategies (Rosa et al., 2020). We need to transform these patents into products by creating new demand. For example, after the implementation of Japan's renewable energy strategy, it achieved constructive influence in the mid-1970s (Ushiyama et al., 1999). Despite these positive results, the scale of manufacturing solar cells is remains small in Japan, and SET innovation and cost reduction must still be solved to promote Japan's renewable energy strategy. These problems should be overcome by utilizing the significant revenue generated by the market. As another example, the Chinese government has taken a series of relevant strategic measures to keep the air clean and achieve energy conservation and emission reduction. Solar energy is widely used in China's urban and rural areas, which benefits millions of people. The rapid development and extensive application of SET in China has improved the quality of China's overall development.

6.2.2. Strengthening basic scientific research

The analysis results of this study show that significant scientific discoveries are undoubtedly the fundamental basis for the emergence and development of SET in different stages. Science advancement pertinent to SET is always carried out in industrial and research laboratories, especially in developed countries. This can also be proved by the retrieval results of SET-related SCI papers from 2010 to 2019. The productive scientific publications largely come from the following nations: China, the United States, India, Germany, England, Spain, Italy, Australia, Japan, and South Korea. The following organizations are prolific in terms of scientific research achievements: Chinese Academy of Sciences, United States Department of Energy, University of California System, Centre National De La Recherche Scientifique, University of Chinese Academy of Sciences, and Indian Institute of Technology System. It is true that the basic scientific progress in China in the last two decades (Zhang et al., 2015), especially in the field of energy-related science, has played a significant role in promoting the development of China's SET (Li et al., 2013; Wu et al., 2014).

Therefore, strengthening basic scientific research remains a crucial strategy for governments around the world in the future. Positive policies can refer to long-term financial support, such as increasing investment in national funds, establishing research teams, and creating national key basic research projects, setting up national science and technology awards, and commending those who have made significant contributions to SET innovation. More importantly, these policies should combine the strengths of industrial labs and universities. For example, China's new "Whole Nation System," which collects different resources from organizations countrywide to overcome scientific difficulties, is a valuable attempt from which other nations and research groups can learn.

6.2.3. Fostering leading companies

Leading companies can transform scientific advancement into more commercial applications and inventions in the market, and boost industry greatly in various areas (Bronfin and Zubarev, 1997), such as IBM in the computer and communication industry, Pfizer in the pharmaceutical industry, Dupont in chemical industry, GE Energy, and Panasonic Corporation in the new energy industry. The SET industry is no exception. The findings of this study show that SET-leading enterprises have emerged in different countries in

various development phases, such as the top 10 SET companies gathered in Japan during 1975–2004. The leading companies vigorously promoted the development of SET. This also applies to latecomers. For example, we can see that the Chinese government has cultivated a group of enterprises in the era of rapid development of SET (Zhao et al., 2017), which will accelerate the development of the SET industry in China. From a global perspective, leading enterprises in SET can also stimulate a wider range of upgrading changes and competitive effects worldwide. However, like other late-developing countries, China lacks leading enterprises with world-class innovation capability in SET, which must be addressed by the government and industry. In the process of transformation, enterprises require more supportive policies from the government, such as encouraging the application of SET in new infrastructure projects, replacing energy-intensive facilities, or implementing more preferential fiscal and tax policies.

6.2.4. Combination of three forces

Our analysis results show that demand, science advancement, and technology catch-up are always the common factors promoting the innovation and development of SET. Therefore, the government needs to implement SET-related policies and give full play to the advantages of the combination of three forces to form a virtuous circle. In this way, it is important, above all, to locate the key forces that the focal country mostly and easily relies on, and take it as the starting point, after which the government may make efforts to invest in the combination of three forces to push SET innovation and industrial development. For example, latecomers may turn largely to technology catch-up forces. They can then follow a circulation path of "technology catch-up-science advancement-demand" in the industry.

6.3. Future directions

There are several interesting future directions elucidated by our findings, such as: How does the theme of SET evolve in a particular country/region? What differences, if any, exist among various regions? Similar research questions may be explored in various institutions, especially for top innovators. More detailed investigations, such as the evolution of solar cells, can be explored from a wide range of perspectives, including related fields, subjects, institutions, and even inventors.

At a macro level, since the evolution of the global SET knowledge network (i.e., the relationship between knowledge elements) can affect the ease of knowledge absorption, transfer, and recombination, future research may build a network based on the common knowledge elements of patents, and study the evolution characteristics of the network. In addition, future studies can construct knowledge transfer networks based on the citation relationship between patents, analyze the evolution characteristics of the global SET innovation citation network, and investigate the impact of geographical proximity and institutional proximity on the evolution. Finally, scholars can also analyze the characteristics of SET innovation evolution from the perspective of technology divergence and convergence.

CRedit authorship contribution statement

Chunjuan Luan: Methodology, Data curation, Visualization, Investigation, Writing - original draft, preparation. **Xiaoming Sun:** Conceptualization, Theory, Writing - review & editing. **Yalan Wang:** Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (NSFC) under Grant numbers 71473028, 71402132, 71774020, 71872141, and 72072140. We are grateful to Zhou Ying from Patsnap Company. The findings and observations contained in this paper are those of the authors and do not necessarily reflect the views of the National Science Foundation of China.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclepro.2020.125019>.

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